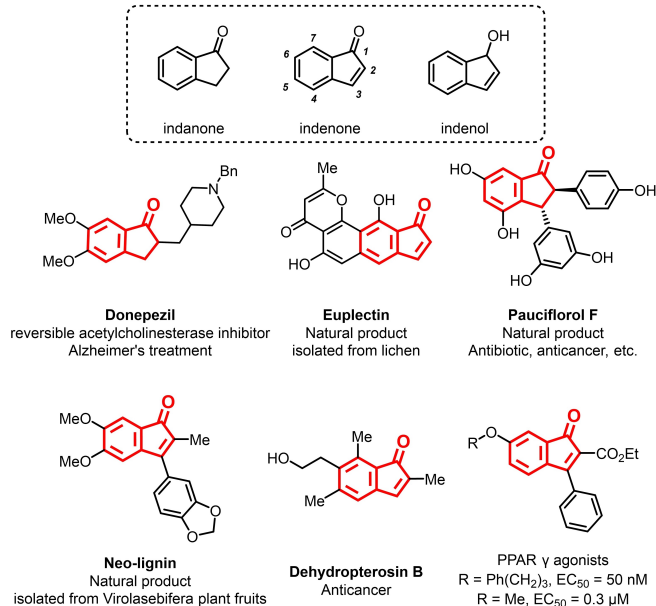


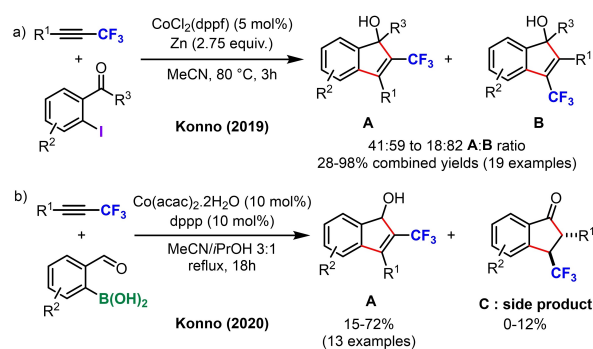
Abstract: A palladium-catalyzed synthesis of 2-SF₅-indenols has been developed by reacting commercially available boronic acid derivatives and readily accessible SF₅-alkynes. The present methodology is fully regioselective thanks to the intrinsic polarization of SF₅-alkynes. A selection of downstream functionalizations has been performed to highlight the versatility of 2-SF₅-indenols and indenones as platforms for the design of more complex SF₅-containing molecules.

Indenone core is a privileged skeleton found in natural products which display relevant bioactive activities such as reversible acetylcholinesterase inhibitor, anticancer, peroxisome proliferator-activated receptor γ (PPAR γ) agonists, among others (Scheme 1.1).^[1] For example, Donepezil, is a FDA-approved drug used as a palliative treatment for Alzheimer's disease^[2] and some natural products like Euplectin or Pauciflorol F exhibit a variety of biological activities such as antibiotic, anticancer, anti-HIV, antioxidant, antifungal, and anti-inflammatory activities respectively.^[3] It is well established that incorporation of a fluorinated motif into organic framework will modify physicochemical properties of molecules, and this strategy is usually used to increase bioavailability, block metabolism, which allows a fine tuning of pharmacokinetic properties.^[4] Nowadays, chemistry of pentafluorosulfanyl group (SF₅) remains a very challenging topic mainly because only few methods are known for introducing SF₅ moiety selectively.^[5–6] SF₅-Alkynes are readily available substrates, easy to prepare from SF₅Cl, and they have been used in heterocyclic synthesis, cycloadditions and hydrofunctionalization reactions.^[7] SF₅Cl is a commercially available gas in Europe, but it could also be prepared from sulfur powder and used as a stock solution in hexane.^[5f–g,6k] Surprisingly, the use of SF₅-alkynes in metal-catalyzed reactions has been poorly explored and is so far limited to gold-catalyzed hydration and hydrofluorination.^[8] To the best of our knowledge, synthesis of SF₅-indenols or indenones is unknown, but analogous reactions with CF₃-alkynes have been reported by the group of Konno.^[9] They developed cobalt-catalyzed methodologies to access either 3-CF₃ (**B**, Scheme 1.2a) or 2-CF₃ indenols (**A**, Scheme 1.2b) depending on the nature of the coupling partner aryl iodide or arylboronic acid, respectively). However, relatively low regioselectivity was obtained in the first case with **A**:**B** ratios between 41:59 and 18:82 (Scheme 1.2a) in favor of 3-CF₃-indenol **B**. In the second case, 2-CF₃-indenol **A** was obtained as major

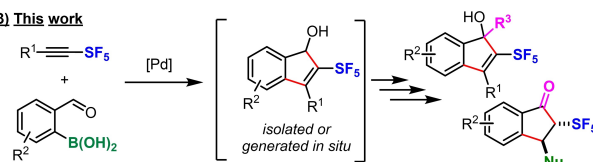
1) Selected indenone or indenone core natural products and bioactive compounds



2) State of the art: CF₃-Indenols



3) This work



Scheme 1. Selected examples of indenones or indenones, state of the art about CF₃-indenols and proposed strategy for regioselective synthesis of 2-SF₅-indenols and further derivatizations.

product in modest to good yields, along with non-negligible amount ($\approx 10\%$) of 3-CF₃-indanone **C** (Scheme 1.2b). Recently, we have reported a fully regio- and stereoselective hydroelementation (N, O, S) of SF₅-alkynes, where we demonstrated that the perfect selectivity was mainly controlled by 1) the polarization of the SF₅-alkyne toward the electron withdrawing SF₅ motif, 2) the bulkiness of the SF₅ moiety and 3) a more accessible LUMO orbital.^[10] In the present report, we decided to explore the reactivity of SF₅-alkynes toward the formation of 2-SF₅-indenols under Pd-catalyzed reaction conditions. Moreover, we examined the synthetic potential of 2-SF₅-indenols as platforms for downstream functionalizations (Scheme 1.3).

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